

CiteView: In-Situ Citation Sensemaking for Scholarly Reading

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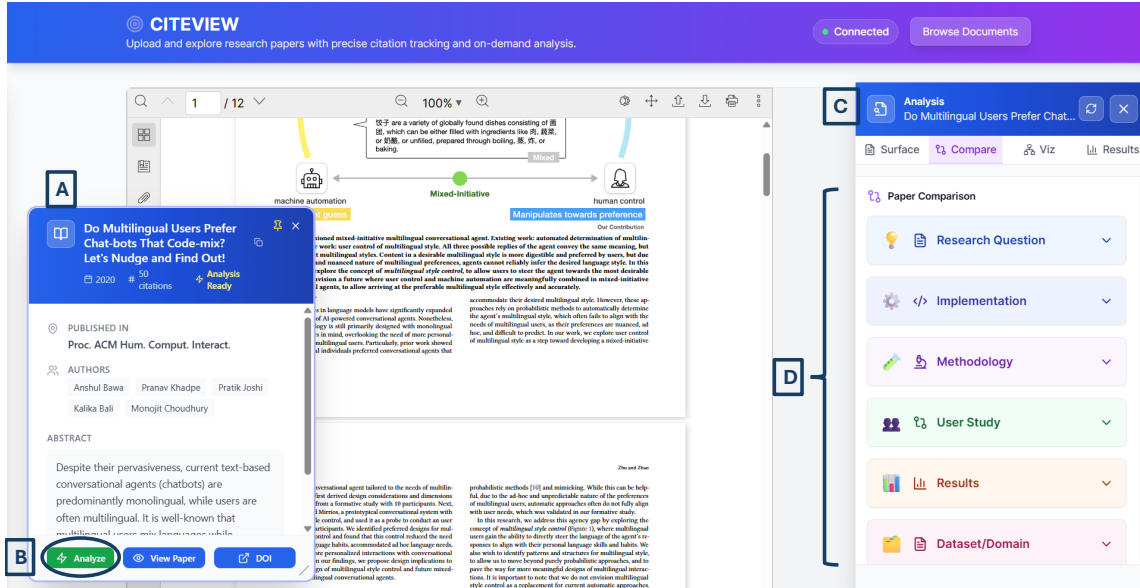


Fig. 1. CiteView interface overview. (A) Paper metadata card displays key bibliographic information. (B) The *Analyze* button initiates on-demand processing of the paper and its citations, triggering extraction of citation contexts, retrieval of cited papers, and generation of multi-modal analysis. (C) The Analysis sidebar opens when users click citations, presenting four complementary tabs: Surface, Compare, Viz, and Results. that respectively surface the most relevant context and citation reasons, contrast the citing and cited papers, visualize the cited work through figures and a semantic graph of its flow, and foreground its main empirical findings.(D) The Compare tab organizes side-by-side comparisons across six research dimensions, including Research Question, Implementation, Methodology, User Study, Results, and Dataset/Domain, each expandable to reveal detailed paired descriptions.

Academic papers contain extensive citations, yet readers often lack the context needed to understand why and how cited works are relevant. We present CiteView, an AI-assisted citation analysis tool that turns inline citations into entry points for contextual understanding. Clicking a citation opens a compact sidebar with layered, citation-aware insights. The sidebar presents contextualized passages from cited papers, citation intent, distilled key results, concise comparisons, and visual links between local argument flow and referenced figures. These capabilities are supported by a multi-agent pipeline for extraction, retrieval, relevance ranking, and synthesis, enabling fast and traceable responses. We evaluated CiteView through a mixed-methods laboratory study with (n=12) participants performing citation-focused reading tasks. We collected interaction logs, post-task questionnaires (SUS and NASA-TLX), and semi-structured interviews. Participants reported positive usability, low workload, and high efficiency. Qualitative feedback

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indicated reduced friction in accessing cited-paper contributions and a preference for insight-driven views focused on highlighting results, enabling direct comparisons, and supporting visual interpretation.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **Interactive systems and tools**; **User interface management systems**.

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1 Introduction

Citations serve as the foundation of scholarly knowledge, linking new contributions to prior work and enabling readers to trace the evolution of ideas across research papers [10, 25]. Yet understanding how a citation supports an argument, whether through empirical evidence, methodological precedent, or contrasting findings, remains surprisingly difficult. Readers encountering a citation face a disruptive choice: accept the author's characterization of the cited work on faith, or break their reading flow to manually verify the claim by navigating to the source paper [17, 23]. This context-switching imposes significant cognitive load [29, 33] and leads readers to either skip verification, risking misinterpretation [2], or invest substantial time investigating references at the expense of deeper analysis [11, 23]. With research papers citing 20-50+ references, this friction makes thorough verification impractical.

Despite citations being fundamental to scholarly argumentation, existing reading tools offer limited support for accessing the specific context needed to interpret them [7, 16, 17].

Existing tools address parts of this problem but not the core challenges, namely the fragmentation resulting from a mismatch between researchers' need to maintain reading continuity and the insufficient support for context-aware citation information revealing citation intent. Text-based assistants like SciSpace Copilot [26] clarify passages but do not reveal how cited papers support claims. Citation platforms such as VOSviewer [31] visualize networks but cannot extract contextually relevant content, figures, results, or evidence from papers themselves. Reference managers like Zotero, Mendeley, and EndNote¹ organize bibliographic metadata but offer limited support for understanding citation intent or retrieving visual evidence."

As a result, researchers must navigate between disconnected systems to access information for a single citation, multiplying cognitive overhead. This reflects a fundamental mismatch: researchers need citation context at the contribution level during reading, but existing tools force them to leave their reading context to find it.

What is needed is a fundamentally different approach: treating citations as integrated knowledge nodes that can be explored in-situ with multi-modal evidence, rather than simple references looked up elsewhere. An effective system should automatically surface what cited papers contribute, why they were cited, and what specific evidence supports claims—all without disrupting reading flow.

To realize this vision, we developed CiteView (Figure 1), the first system to integrate multi-modal citation analysis, textual summaries, visual evidence extraction, quantitative findings, and cross-paper comparison, directly within the PDF reading interface. The system introduces three key innovations: (1) *multi-modal evidence extraction* that automatically surfaces figures, tables, and results alongside text; (2) *contribution-focused analysis* that identifies what specific contributions cited papers make through purpose classification and structured comparison across six research

¹Zotero: <https://www.zotero.org/>; Mendeley: <https://www.mendeley.com/>; EndNote: <https://endnote.com/>

dimensions; and (3) *on-demand processing* that works with arbitrary user-uploaded PDFs without requiring pre-indexed databases.

CiteView addresses three core challenges in citation verification: (1) *eliminating context switching* by surfacing all citation-relevant information, text, figures, results, and relationships, directly in the reading interface; (2) *revealing citation intent* through purpose classification and cross-paper comparison, which shows what cited papers contribute and why they were cited; and (3) *enabling on-demand analysis* of any user-uploaded PDF without requiring pre-indexed databases.

When users upload a research paper, CiteView displays metadata (Figure 1A) and enables citation analysis via the Analyze button (Figure 1B). Clicking a citation opens an Analysis sidebar (Figure 1C) with four tabs: *Surface* (context-aware explanations and citation purpose), *Compare* (Figure 1D, side-by-side analysis across Research Question, Methodology, Implementation, User Study, Results, Dataset/Domain), *Viz* (semantic graphs and extracted figures), and *Results* (quantitative and qualitative findings). This integrated interface enables exploration without breaking reading flow.

We evaluated CiteView in a mixed-methods study involving 12 participants who completed citation-focused reading tasks. Results demonstrate effectiveness: participants achieved near, perfect task completion (97.9%) with remarkably low cognitive load (Mental Demand: $M=2.0/10$; Effort: $M=2.1/10$), substantially lower than typical academic reading tasks. Usability scores (SUS: $M=71.9$) exceeded industry acceptability thresholds, while ratings for time efficiency ($M=8.9/10$) and ease of finding information ($M=9.1/10$) indicate CiteView successfully addresses core friction points in citation exploration. Critically, qualitative feedback revealed that CiteView fundamentally transformed participants' citation exploration experience. Participants reported that the system "transforms citation checking from a tedious interruption into a fluid part of reading," enabling them to verify claims they would typically skip due to time constraints. Participants particularly valued insight-driven views that directly revealed what cited papers contributed, why they were cited, and what specific evidence supported the claims being made.

This paper makes three primary contributions:

(1) CiteView, the first system to integrate multi-modal, contribution-focused citation analysis directly within the PDF reading interface. Unlike prior work that treats citations as simple references or provides generic summaries, CiteView reveals what specific contributions cited papers make and why they were cited through integrated textual analysis, visual evidence extraction, and cross-paper comparison, enabling comprehensive verification without context switching.

(2) Design insights and an interaction model for contribution-focused citation support. We introduce a four-view interaction model that prioritizes what cited papers contribute over what they contain, and identify key principles: readers need contribution-level understanding, multi-modal evidence is essential for verification, and cross-paper comparison enables deeper sensemaking. These contributions provide a framework for future intelligent reading interfaces.

(3) Empirical evidence that contribution-level citation context fundamentally improves scholarly reading. Our mixed-methods study ($n=12$) demonstrates that multi-modal, in-situ citation analysis achieves near-perfect task completion (97.9%) with low cognitive load (Mental Demand: $M=2.0/10$), while participants strongly preferred insight-driven views for understanding citation relevance. These findings validate the contribution-focused approach and inform the design of future citation support tools.

Together, these contributions advance the design of intelligent reading tools that make scholarly exploration more fluid, accessible, and cognitively manageable.

2 Related Works

Our work builds upon two intersecting areas: interactive tools for scholarly reading and citation analysis, and AI-assisted document understanding. Despite significant progress in each domain, there remains a lack of integrated support for contextual citation analysis during active reading.

2.1 Interactive Reading Tools and Citation Network Systems

Prior research on scholarly reading has consistently shown that academic reading is not a purely linear activity; instead, it involves iterative navigation, annotation, and cross-referencing as readers interpret and organize ideas [14]. Early systems like XLibris and LiquidText [27, 30] supported active reading through flexible navigation and spatial reorganization. Building on this, recent tools augment PDFs to reduce reading friction: ScholarPhi [9] surfaces just-in-time definitions, Paper Plain [3] adds plain-language summaries for non-experts, and Scim and PaperMage [7, 17] highlight key sentences and reflow layouts. While effective for local comprehension, these tools treat PDFs as isolated artifacts rather than entry points into broader evidence networks.

Research on sensemaking emphasizes that comprehension requires synthesizing relationships between a paper’s content and external scholarly knowledge [19, 21]. Some recent systems integrate citation context into reading: PaperQuest [22] suggests relevant papers during literature review, CiteSee [5] augments inline citations with personalized relevance indicators, and CiteRead [23] annotates documents with commentary from subsequent works. Related efforts explore the discovery of the literature through author networks: ComLittee [12] enables scholars to curate committees of key authors to guide the recommendations of the articles, while Synergi [13] explores mixed-initiative workflows to synthesize research threads in multiple articles, helping scholars build hierarchical structures of related work. However, these systems primarily focus on incoming citations, author-based discovery, or text-based synthesis rather than analyzing outgoing references, extracting visual evidence, or enabling in-situ verification of specific cited claims within the active reading flow.

To navigate literature growth, numerous systems visualize citation networks and bibliometric patterns. Traditional tools like CiteSpace and VOSviewer [6, 31, 32] provide bibliometric overviews for identifying clusters and trends. Interactive platforms like Connected Papers, ResearchRabbit, Inciteful, and Litmaps² explore citation neighborhoods through co-citation relationships. While effective for discovery and landscape mapping, they operate exclusively on metadata and cannot surface specific content or figures that make citations relevant. These systems operate at the macro-level of citation networks rather than the micro-level of individual claims, showing connections between papers without explaining how specific cited content supports particular arguments. Most depend on pre-indexed databases [1] and cannot analyze user-provided PDFs on demand. Collectively, these tools leave readers without integrated support for verifying individual citations through direct access to the specific findings, figures, and contextual evidence that justify cited claims within the active reading flow.

2.2 AI-Assisted Document Understanding

Large language models have enabled AI-powered reading assistants that help researchers understand academic papers. Systems like SciSpace [26], Elicit [8], and Consensus [20] use generative AI to summarize content and answer questions. Smart citation platforms like scite.ai [24] classify citation statements as supporting, contrasting, or neutral. However,

²<https://www.connectedpapers.com/>, <https://www.researchrabbit.ai/>, <https://inciteful.xyz/>, <https://www.litmaps.com/>

these tools typically treat documents as unstructured text and overlook visual elements like figures and tables, resulting in outputs that lack specificity for understanding how cited works support particular claims [15].

The Semantic Reader project integrates AI-driven context into reading interfaces by analyzing document structure [17]. Features like inline Citation Cards present key details at the point of citation, while CiteRead and CiteSee provide in-situ annotations [5, 23]. Despite this progress, these systems rely on indexed corpora, cannot process arbitrary user-uploaded PDFs, and offer limited support for extracting figure-level evidence, primarily facilitating textual understanding without integrated multimodal support for citation intent [17, 24].

Inspired by CitePeek’s approach to surfacing contextual paragraphs through similarity scoring [4], CiteView extends in-document citation analysis by integrating comparative, visual, and multi-dimensional analytical capabilities. This reflects the recognition that citation comprehension requires understanding not only what a cited work contributes, but also how it relates to the current paper, what visual evidence it provides, and how its findings connect across analytical dimensions.

In summary, prior work has advanced important components of the scholarly workflow: active reading tools support annotation, citation network systems enable discovery, and AI-driven services provide summarization and semantic insight [6, 18, 28, 31, 32]. However, these capabilities remain fragmented. Reading tools rarely integrate external sources, visualization platforms operate away from the moment of reading, and AI assistants function outside the natural flow of PDF-based engagement. Even systems that incorporate limited context, such as CiteSee or CiteRead do not deliver concrete evidence from cited papers directly into the reading view.

This fragmentation motivates the need for an integrated, in-document approach. CiteView addresses this gap by treating the PDF as an entry point to a connected scholarly knowledge graph rather than a static endpoint. Unlike tools that rely on context switching or pre-indexed databases [6, 17, 24], CiteView operates on-demand on arbitrary PDFs to surface citation intent, comparative relationships, and visual evidence from referenced figures or tables directly alongside the cited text, enabling efficient verification and deeper cross-document sensemaking without disrupting the reader’s flow.

3 System Overview

CiteView (Figure 1) transforms static PDFs into citation, centric exploration environments. When users upload a paper and click citations, a sidebar opens with four views (Surface, Compare, Viz, Results) providing contextual summaries, cross-paper comparisons, visualizations, and key findings directly within the reading interface.

3.1 Surface Tab

The Surface tab is the default view that opens when a reader clicks a citation, providing a fast, context-aware explanation of what the cited paper contributes in that specific place in the source document. It is designed not to summarize the entire article, but to answer a narrower question: “Why is this paper being cited here, and what does it add?” To achieve this, it combines the local citation context in the source PDF with selectively extracted content from the cited paper and transforms it into three to four concise, evidence-grounded bullet points that can be read at a glance. (Figure 2)

When a user clicks an inline citation, the system captures the complete sentence containing the citation (typically 15-40 words) and treats it as a query encoding the topic and rhetorical role of the citation (e.g., introducing a method, supporting a result). The cited paper is processed using GROBID³ for metadata extraction and PyMuPDF for full-text

³GROBID: <https://github.com/kermitt2/grobid>

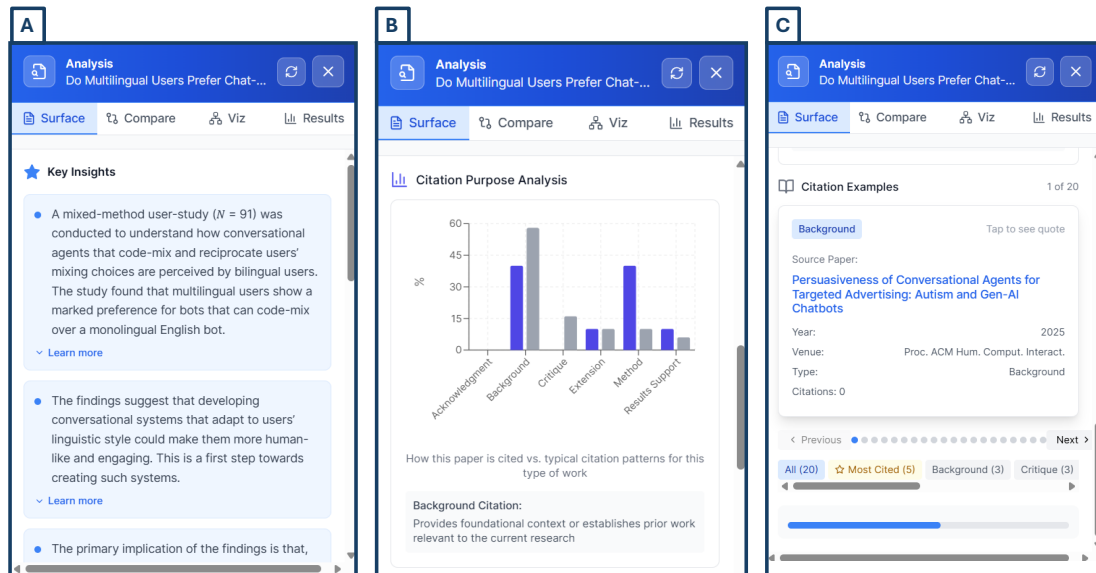


Fig. 2. Surface tab views in CiteView. (A) shows context-aware bullet points that explain the cited paper's role in the current citation, (B) visualizes citation-purpose distributions, and (C) lists example citing papers that use the same work, each card revealing the exact quotation when clicked.

extraction. Text is segmented into sentences, which are then ranked by relevance using a multi-component algorithm combining semantic similarity, keyword overlap, and section importance. This approach captures both lexical and semantic relationships while prioritizing content from methodologically important sections (e.g., Methods, Results). The top-ranked sentences (typically 5-8) are selected for analysis. The citation context and top-ranked sentences (300-600 words) are passed to OpenAI's GPT-4⁴. Structured prompts instruct the model to generate exactly 4 bullet points that explain the cited work's specific role and tie it to the local context, emphasizing concrete methods or results rather than generic summaries.

In the interface, the Surface tab presents these bullets alongside the original citation context, allowing readers to see both the author's sentence and the system's interpretation in one place (Figure 2.A). Each bullet is initially displayed in a compact form, with controls that allow users to expand it for full text when more detail is needed, supporting both quick scanning and in-depth inspection.

The Surface view displays citation-purpose distributions indicating how the work is typically used in the literature, such as background, methodological support, or results support, based on Semantic Scholar's citation-intent labels (Figure 2.B). These aggregated distributions are presented in a chart, revealing patterns such as a paper being cited primarily for its methods and rarely for its critique. The view also presents example citations from other papers, each annotated with citation-purpose labels and basic metadata (title, venue, year). Selecting a citation card reveals the exact quoted passage, enabling readers to quickly understand how the cited paper is used in context, whether as background, methodology, or comparative results (Figure 2.C).

⁴OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

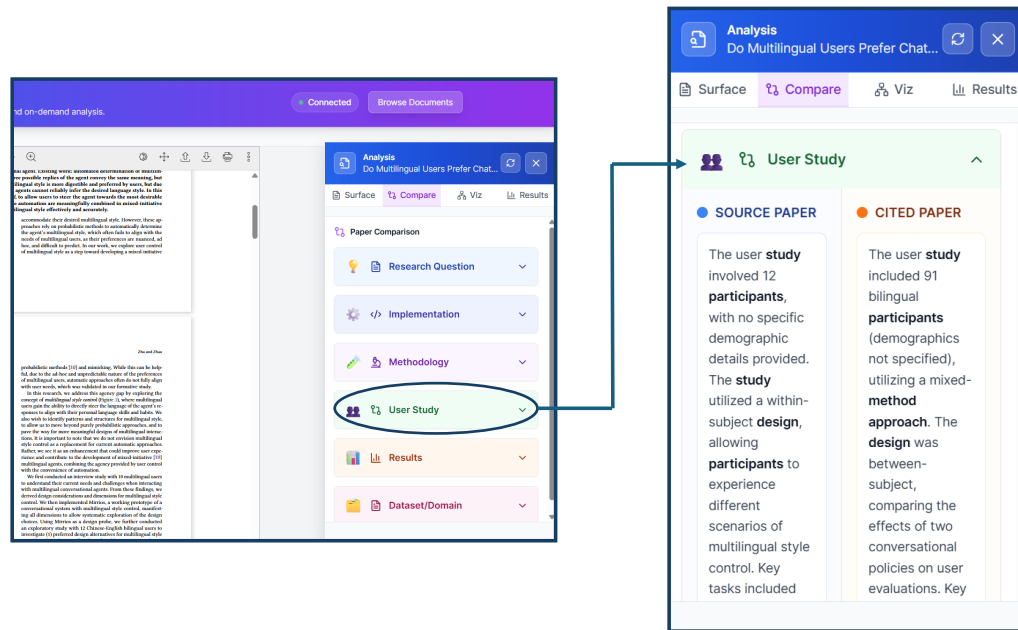


Fig. 3. Compare tab in CiteView. The left panel shows the main PDF reader with the Compare tab selected, and the zoomed view highlights a user-study card comparing the source paper (left column) and the cited paper (right column) along participants, design, and other study details.

3.2 Compare Tab

The Compare tab enables readers to understand how a cited work relates to the source paper across six research dimensions: research question, methodology, implementation, user study, results, and dataset/domain (Figure 3).

These six dimensions address the core questions readers typically ask when evaluating cited work: what problem does it address (research question), how does it approach the problem (methodology), what system was built (implementation), how was it evaluated (user study), what was found (results), and where does it apply (dataset/domain). This selection emerged from iterative design reflecting how researchers structure empirical work in HCI, spanning the complete arc from problem framing through solution development to validation and contextualization. The dimensions balance comprehensive coverage with cognitive manageability. We limited the structure to six rather than more granular categories based on pilot testing with two graduate students, which indicated that more than six dimensions fragmented comparisons into overly specific units while requiring excessive scrolling.

When content is missing for a dimension (e.g., a theoretical paper lacks implementation, or a survey paper has no original user study), the system explicitly indicates absence rather than generating speculative content, displaying messages such as "Implementation not available, theoretical paper" or "User study not conducted, evaluation based on benchmarks." When dimensions are fundamentally incomparable (e.g., comparing a survey paper's literature review methodology with an experimental paper's controlled study), each paper's approach is described independently to avoid misleading parallels.

When a reader opens the Compare tab, the system employs a two-agent workflow built with the CrewAI framework⁵. An Ingestion Agent first extracts structured content from both papers by identifying relevant sections (e.g., "Methods," "Results," "Implementation") through header detection and keyword matching, then capturing key information such as technical terms, system components, evaluation metrics, and experimental designs. This agent utilizes GPT-4⁶ with dimension-specific prompts that specify the information to extract for each comparison dimension. An Analyst Agent then receives this structured content and generates six paired descriptions—one summarizing the source paper's approach, one for the cited paper—written in plain language with concrete details such as "uses BERT embeddings with fine-tuning on 10,000 papers" rather than generic statements like "uses modern NLP techniques."

Comparisons are displayed as collapsible side-by-side cards (source left, cited right), allowing readers to expand specific dimensions for detailed examination or scan across all six at once. This design transforms the manual process of switching between PDFs into a structured comparative overview that supports higher-level reasoning about the literature, such as identifying complementary methods, spotting opportunities for extension, or detecting when citations are methodologically misaligned with the work that cites them.

3.3 Viz Tab

The Visualization tab addresses a fundamental challenge in citation analysis: readers often need to quickly verify specific claims or understand a paper's core contribution without reading the entire document. When evaluating whether a cited paper is relevant, readers must determine what problem it addresses, what approach it proposes, and what results it achieves. Information is scattered across multiple sections. The tab provides two complementary visual representations. The Graph sub-tab displays key concepts as an interactive node-link diagram, where nodes represent research elements (problems, approaches, findings). This enables readers to quickly grasp the paper's conceptual structure and contribution flow. For example, understanding how a specific problem motivated a particular approach that produced certain findings. The Figure sub-tab extracts and displays key figures from the cited paper with their captions, providing direct access to visual evidence without opening the cited PDF. Together, these views move beyond linear text summaries by exposing the conceptual structure of a paper and its visual evidence, enabling readers to reason about cited work more efficiently within the reading context.

3.3.1 Graph.

The Graph view generates a compact semantic knowledge graph capturing the cited paper's argumentative structure (Figure 4). CiteView extracts and segments the paper's text, then processes each section using GPT-4⁷ with section-specific prompts to extract concise statements as typed nodes. We selected eight node types, Problem, Approach, System, Finding, Dataset, Insight, Background, and Critique, to cover the essential elements of research contributions while maintaining visual clarity. These categories emerged from an analysis of how authors structure arguments in empirical papers, including identifying problems, proposing approaches, implementing systems, reporting findings, describing datasets, offering insights, providing background, and critiquing limitations. Nodes are connected by typed edges (e.g., "motivates," "implements," "produces") inferred through two complementary methods: (1) rule-based heuristics that apply predefined relationship patterns (e.g., Problem nodes connect to Approach nodes with "motivates" edges, Approach nodes connect to System nodes with "implements" edges, System nodes connect to Finding nodes with "produces" edges), and (2) semantic similarity scoring between node text pairs nodes with cosine similarity above 0.6

⁵CrewAI: <https://www.crewai.com/>

⁶OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

⁷OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

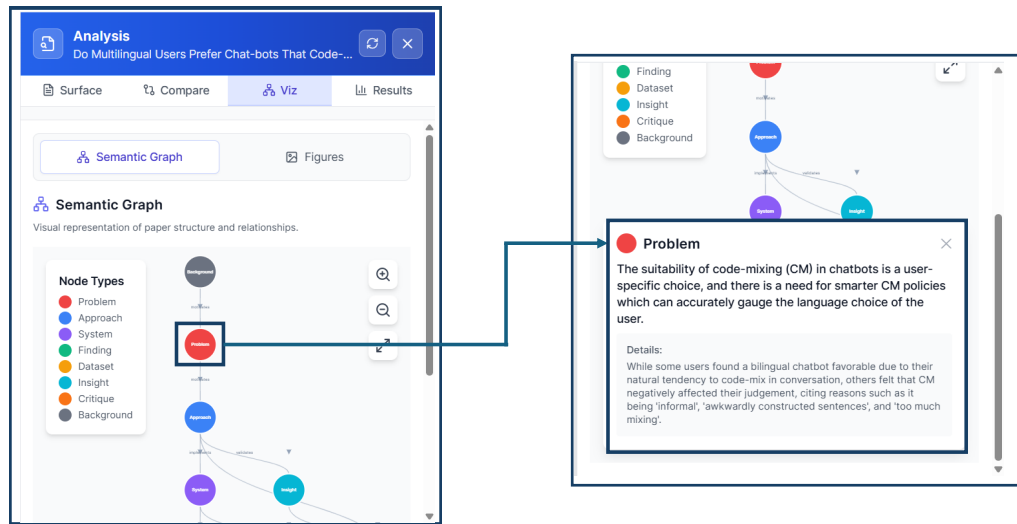


Fig. 4. Semantic Graph view showing the conceptual structure of a cited paper. Clicking on a node (e.g., Problem or Approach) opens a detailed explanation, supporting overview-to-detail exploration without leaving the reading context.

threshold receive "relates to" edges. This hybrid approach balances structural precision (rule-based) with flexibility to capture unexpected relationships (similarity-based). When readers click a node, CiteView displays a focused textual explanation with supporting details, allowing for an overview-to-detail exploration without requiring linear reading.

3.3.2 Figure.

The Figure view surfaces three curated figures from the cited paper: a teaser figure providing a high-level overview, the most important figure reflecting the paper's primary contribution, and the most related figure relevant to the current citation context (Figure 5). This three-figure selection emerged from balancing competing needs: readers require both broad understanding (teaser, important) and citation-specific evidence (relevant), but showing all figures (often 6-10 per paper) would overwhelm the interface and require excessive scrolling. Pilot testing with two graduate students confirmed that three figures provided adequate coverage—participants could locate relevant visual evidence without being overwhelmed—while showing only one figure frequently missed key contributions. CiteView extracts figures using PyMuPDF⁸ to detect embedded image objects and identifies captions by searching for figure references (e.g., "Figure 1") within proximity thresholds of image boundaries. Figures are ranked by: (1) position in document (earlier figures prioritized), and (2) semantic similarity between figure caption and citation context (computed using sentence embeddings). The top-ranked figure from each category (position-based, context-relevance) is selected to provide diverse coverage.

⁸PyMuPDF: <https://pymupdf.readthedocs.io/>

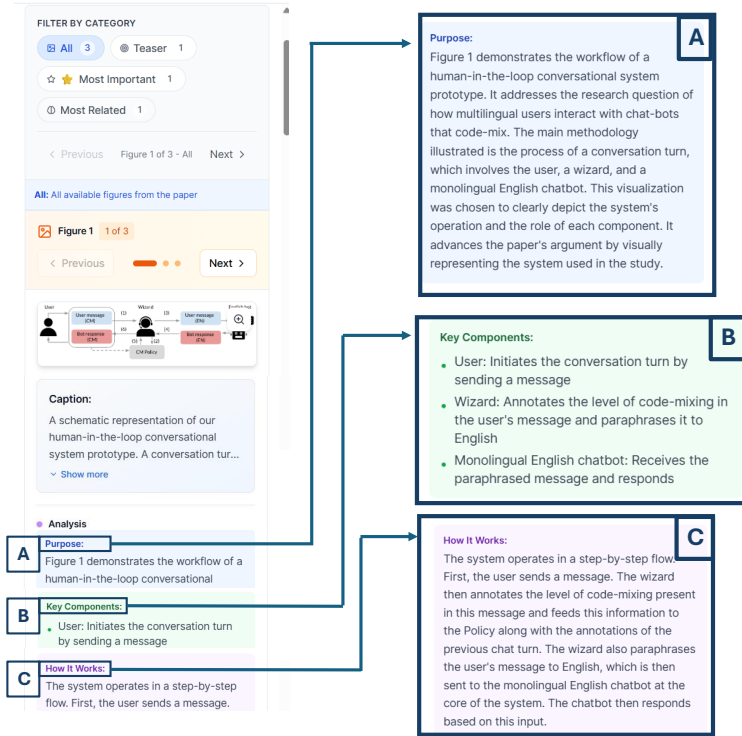


Fig. 5. Figure view in CiteView showing an extracted figure from a cited paper with structured explanations. (A) describes the figure's purpose and research question, (B) lists key components and their functions, and (C) provides a step-by-step explanation of how the depicted system works. Navigation controls allow browsing multiple figures (1 of 3 shown).

Each figure is accompanied by a three-part explanation generated by GPT-4⁹. We designed this structure to address the challenge that figures in academic papers are often difficult to interpret without reading surrounding context. Readers may see a system diagram or results graph but not understand its significance or how to read it. The three-part format (A) Purpose describes the figure's role and research question (Figure 5A), (B) Key Components lists main visual elements (Figure 5B), and (C) How It Works explains how elements interact (Figure 5C)—follows cognitive principles of figure comprehension (readers need purpose, then components, then mechanism). The prompt instructs GPT-4¹⁰ to analyze the figure caption and surrounding text, generating these sections with specific constraints: Purpose (2-3 sentences), Key Components (3-5 bulleted items), and How It Works (sequential explanation). This enables figure interpretation without opening the cited PDF.

3.4 Result Tab

The Results tab addresses a fundamental challenge in understanding cited papers: readers often need to quickly grasp what a paper actually found the key performance metrics, user study outcomes, or empirical insights, to assess the paper's contributions and determine its relevance to their work. Manually extracting results requires navigating dense

⁹OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

¹⁰OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

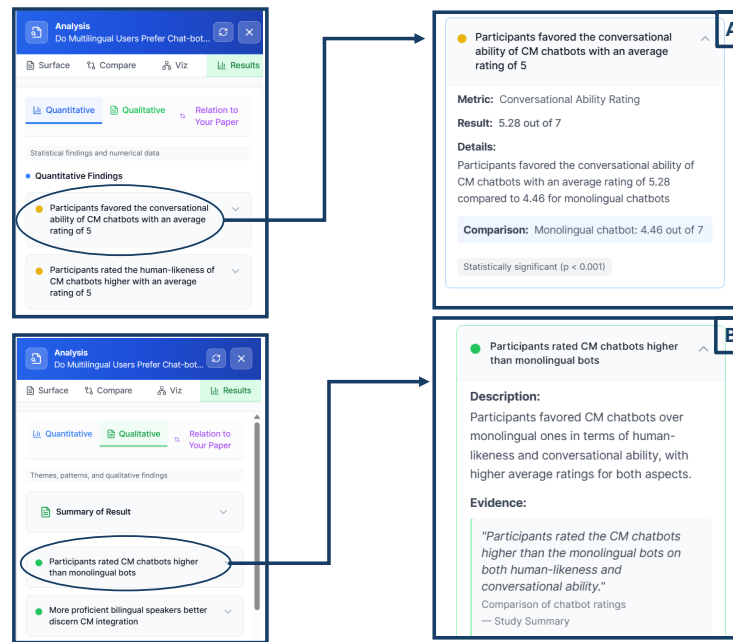


Fig. 6. Results tab showing quantitative, qualitative, and relation views that summarize cited paper findings and explain their relevance to the current paper, supporting efficient interpretation without leaving the reading context. (A) presents quantitative outcomes as structured units with metrics, values, and statistical significance, (B) surfaces qualitative themes and insights with supporting evidence.

sections of results that contain multiple tables, figures, and scattered findings throughout the text. The Results tab automates this by extracting and structuring findings through three complementary views. These three views cover the essential types of information readers need from results sections: what was measured (quantitative), what was observed (qualitative), and why it matters to the current citation (relation). The Quantitative view extracts numerical outcomes as structured units containing measured metrics (e.g., "accuracy"), reported value (e.g., "85%"), comparison contexts (e.g., "vs. baseline: 78%"), and statistical significance (e.g., " $p < 0.05$ "), enabling readers to quickly understand performance achievements. The Qualitative view identifies non-numerical findings such as user behaviors, interview themes, or design insights, presenting each as a concise claim paired with supporting evidence from the paper. This addresses the challenge that qualitative findings are scattered throughout results and discussion sections without standardized formats. The Relation view explains how cited results connect to the source paper through four types: validation (cited results support source claims), comparison (cited results provide a performance baseline), extension (source builds upon cited results), and critique (source identifies limitations in cited results). Content is extracted using three specialized CrewAI¹¹ agents, each equipped with GPT-4¹² and domain-specific prompts: the Quantitative Analysis Agent for numerical findings, the Qualitative Analysis Agent for themes and insights, and the Relation Analysis Agent for cross-paper connections. Together, these three views provide comprehensive coverage of empirical findings, enabling readers to rapidly assess a cited paper's contributions and relevance without manually searching through dense results sections.

¹¹CrewAI: <https://www.crewai.com/>

¹²OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

4 System Implementation

CiteView is implemented as a full-stack web system integrating structured document processing, interactive PDF rendering, external scholarly metadata services, and LLM-based analysis. The frontend is built with React and TypeScript; the backend uses Python with Flask. The system combines rule-based extraction methods with LLM-based synthesis to balance reliability and expressiveness.

PDFs are rendered using PDF.js. Citations are detected through GROBID (TEI-XML structure) and PyMuPDF (coordinate extraction).

Metadata is enriched via Semantic Scholar with SQLite caching. CiteView queries open-access endpoints in sequence: Semantic Scholar, Unpaywall¹³, arXiv¹⁴, and PubMed Central¹⁵. Retrieved PDFs are validated and cached. If no accessible PDF documents are found, the system degrades to metadata-only analysis.

CiteView employs retrieval-augmented generation (RAG): relevant content is extracted from cited papers, then provided to GPT-4¹⁶ for grounded analysis. Text is segmented into sentences and ranked using weighted multi-component scoring combining semantic similarity (sentence-transformers with all-MiniLM-L6-v2), keyword overlap, section importance, and position. The top 8 sentences are selected and passed to GPT-4¹⁷ with structured prompts. For Surface analysis, prompts generate 4 bullet points (15-25 words each) emphasizing concrete methods or results. For the Compare and Results tabs, prompts enforce structured JSON output. To minimize hallucination, outputs are constrained through evidence-grounding (operating only on retrieved sentences), structured formats, external validation (Semantic Scholar citation labels), explicit refusal instructions, and rule-based fallbacks.

Compare and Results tabs employ a multi-agent system built with CrewAI¹⁸. The Compare tab uses two agents: an Ingestion Agent that extracts structured content from both papers, and an Analyst Agent that generates six paired descriptions. The Results tab uses three specialized agents for quantitative, qualitative, and relational analysis. All agents use GPT-4 with task-specific prompts and output structured JSON.

Figures are extracted using PyMuPDF¹⁹ and ranked by position, size, and semantic similarity to citation context. Each figure includes a GPT-4-generated explanation with three parts: Purpose, Key Components, and How It Works. Semantic graphs extract typed statements (Problem, Approach, System, Finding, Dataset, Insight, Background, Critique) using GPT-4, limited to 5-10 nodes. Relationships are inferred through rule-based heuristics and semantic similarity, rendered using D3.js²⁰ force-directed layout.

5 User Study

To evaluate how CiteView supports citation sensemaking during academic reading, we conducted a user study examining its impact on reading flow, citation comprehension, and users' ability to interpret cited work without leaving the source paper.

¹³Unpaywall: <https://unpaywall.org/>

¹⁴arXiv: <https://arxiv.org/>

¹⁵PubMed Central: <https://www.ncbi.nlm.nih.gov/pmc/>

¹⁶OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

¹⁷OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

¹⁸CrewAI: <https://www.crewai.com/>

¹⁹PyMuPDF: <https://pymupdf.readthedocs.io/>

²⁰D3.js: <https://d3js.org/>

5.1 Participants

We recruited 12 participants (4 women, 8 men) aged 23-26 ($M=24.5$), including 10 master's students and 2 senior undergraduates. Ten were from computer science, and two were from physics and management. All noted that they routinely read academic papers as part of their research or coursework and rely on citations to interpret related work. Participants were recruited through email invitations sent to university mailing lists. As an incentive, three participants were randomly selected to receive a \$20 gift card for their time.

5.2 Study Design and Procedure

After reading the consent form and receiving an overview of the study, we introduced CiteView in detail, describing the system's purpose and the function of each tab (Surface, Compare, Results, and Visualization). Participants were then given time to explore the interface and interaction flow using a sample research paper. Once they were familiar and confident with the system, we briefed them on the tasks they would complete.

We selected three recent HCI research papers that cited multiple related works, each containing 30-50 citations with diverse citation purposes (background, methodology, results support). To test CiteView's reliability across different paper types, we employed a between-subjects design: participants were randomly divided into three groups of four, with each group working with a different source paper throughout all three task blocks. This design ensured that our findings reflected CiteView's performance across diverse papers rather than being specific to a single document, demonstrating the system's generalizability across different HCI research contexts. For each paper, we pre-selected specific citations that were well-supported by CiteView's analysis capabilities, meaning the cited papers were openly accessible, contained clear empirical results, included figures, and could be meaningfully compared to the source paper across multiple dimensions. This selection ensured that all tabs would contain substantive information for participants to explore.

Each participant completed three task blocks designed to gradually build familiarity with CiteView. In Block 1, participants were assigned a source paper and instructed to click on a specific citation (e.g., "[15]") within that paper. They were then explicitly told to use only the Surface and Compare tabs to answer four multiple-choice questions, two questions per tab. Questions for the Surface tab focused on citation context and purpose (e.g., "According to the Surface view, why is this paper cited in this context?" and "How is this work typically cited in the literature?"). Questions for the Compare tab focused on cross-paper relationships (e.g., "How does the cited paper's user study design differ from the source paper's approach?" and "Which research question dimension shows the greatest similarity between the two papers?"). platform's overall usability: (1) it introduced participants to CiteView's ability to surface citation-specific context (Surface) and comparative analysis (Compare), and (2) it established that different tabs serve different analytical purposes. By restricting tab usage, we ensured participants engaged deeply with these two views before moving to visualization-oriented tabs.

Block 2 used the same citation from Block 1 but required participants to switch to the Results and Visualization tabs to answer four new multiple-choice questions, again, two per tab. Questions for the Results tab focused on empirical findings (e.g., "What were the key quantitative results reported in the cited paper?" and "What qualitative themes emerged from the user study?"). Questions for the Visualization tab focused on visual evidence and paper structure (e.g., "What does the main figure in the cited paper illustrate?" and "According to the semantic graph, what problem does the cited paper address?").

By reusing the same citation, participants could compare how Surface/Compare views (Block 1) complemented Results/Visualization views (Block 2) for understanding a single cited work. This design reinforced that CiteView provides multi-faceted analysis rather than redundant summaries. After completing Block 2, participants had now used all four tabs in a guided manner.

Block 3 assessed whether participants could independently select appropriate tabs to answer citation questions. Participants were assigned a different citation from the same source paper and given four multiple-choice questions without any restrictions on tab usage. Questions were intentionally varied to require different types of information (e.g., one question might require Results data, another might require Compare analysis, and another might require Visualization evidence). Participants had to determine which tabs contained the necessary information and navigate accordingly. This block served as an ecological validity check: after guided exposure in Blocks 1-2, could participants use CiteView effectively in a more realistic scenario where they must decide how to explore citations on their own? We recorded which tabs participants used for each question through interaction logs, allowing us to observe naturalistic tab selection patterns.

All multiple-choice questions were generated manually by the research team through a two-step process. First, for each pre-selected citation, we used CiteView to analyze the cited paper and extract information across all four tabs. Second, we formulated questions whose answers could be directly verified from the displayed content (e.g., if the Compare tab showed "User Study: Source paper used 12 participants; Cited paper used 8 participants," we created a question like "How many participants were in the cited paper's user study?"). This ensured questions were answerable using CiteView without requiring external knowledge or subjective interpretation. Each question was piloted during our preliminary testing to confirm clarity and answer correctness.

After completing all three task blocks, participants filled out the NASA-TLX workload questionnaire and the System Usability Scale (SUS). They then participated in a 20-minute semi-structured interview where we asked them to reflect on their experience, describe which features they found most useful, identify any points of confusion, and compare CiteView to their typical citation-following workflows. The entire study session lasted approximately 60 minutes. The procedure was validated through pilot testing with two graduate students, resulting in minor clarifications to task instructions.

5.3 Measures and Data Collection

We collected task performance (percentage correct), system logs (time per tab), NASA-TLX workload ratings, SUS usability scores, and semi-structured interview data. Quantitative measures were analyzed using descriptive statistics, while interview transcripts were thematically coded to identify recurring patterns.

6 Results

We report results from a mixed-methods user study combining interaction logs, standardized questionnaires (NASA-TLX and SUS), and post-task semi-structured interviews. Overall, participants were able to complete citation-focused tasks using CiteView and reported positive experiences with the system. Usability scores were high and perceived workload was generally low, while interview feedback highlighted both the benefits of inline citation exploration and areas for improvement. The following sections detail task performance, questionnaire results, and recurring themes from participant feedback.

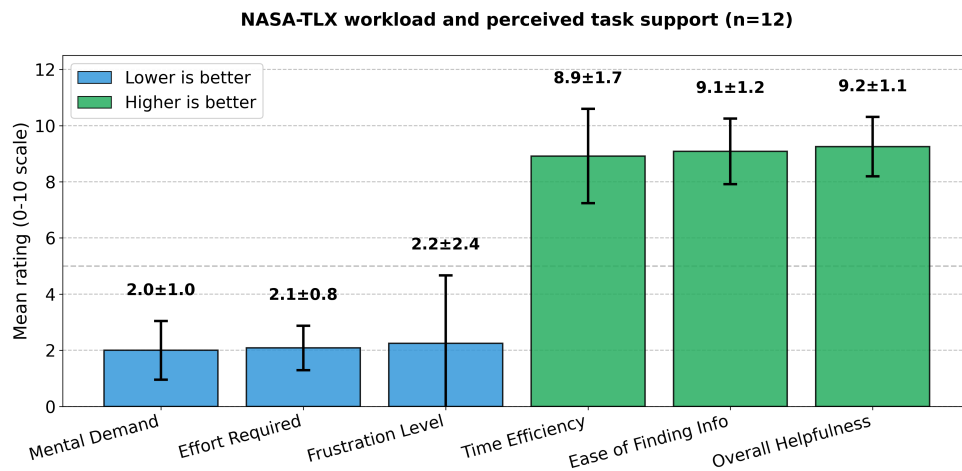


Fig. 7. NASA-TLX workload and perceived task support ratings ($n=12$, 0-10 scale). Error bars represent standard deviation. Lower scores indicate reduced cognitive and physical workload; higher scores indicate better efficiency and support. CiteView demonstrated low workload demands and high perceived usefulness across all dimensions.

6.1 Task Performance

Across all study sessions, participants were generally able to answer the citation-focused questions correctly using CiteView. Most participants answered all questions correctly for the majority of citations, with occasional skipped or incorrect responses limited to one or two questions per task. Errors and omissions were typically associated with ambiguity in the question wording or uncertainty about which aspect of the cited paper to prioritize, rather than difficulty operating the interface itself (e.g., uncertainty about what counted as the “main figure”). Overall, across $n = 12$ participants (12 questions each), the mean number of correctly answered questions was 11.75/12 ($SD=0.62$; $Mdn=12$; range: 10–12), corresponding to a 97.9% completion rate. Ten participants (83.3%) achieved perfect performance (12/12), and 11/12 participants (91.7%) answered at least 11/12 questions.

6.2 Usability and Workload

SUS results indicated generally positive usability ($M=71.9$, $SD=13.2$, $n=12$), above the industry benchmark (68) and approaching the “good usability” threshold (73.5). Item-level responses showed clear polarity: participants rated positively worded items high (intention to use, ease of use, feature integration, learnability, confidence) while rating negatively worded items low (unnecessary complexity, need for support, inconsistency, cumbersomeness). This pattern suggests CiteView was experienced as learnable and coherent, with remaining friction related to interface organization rather than basic operability (Figure 8).

NASA-TLX ratings indicated efficient citation exploration with low perceived effort. Participants reported low cognitive and physical burden (Mental Demand: $M=2.0$, $SD=1.0$; Effort Required: $M=2.1$, $SD=0.8$ on 0-10 scale), while giving consistently high ratings for speed and information access (Time Efficiency: $M=8.9$, $SD=1.7$; Ease of Finding Information: $M=9.1$, $SD=1.2$; Overall Helpfulness: $M=9.2$, $SD=1.1$). Frustration scores were more mixed ($M=2.2$, $SD=2.4$), suggesting some participants experienced friction with interface organization or visualization clarity (Figure 7).

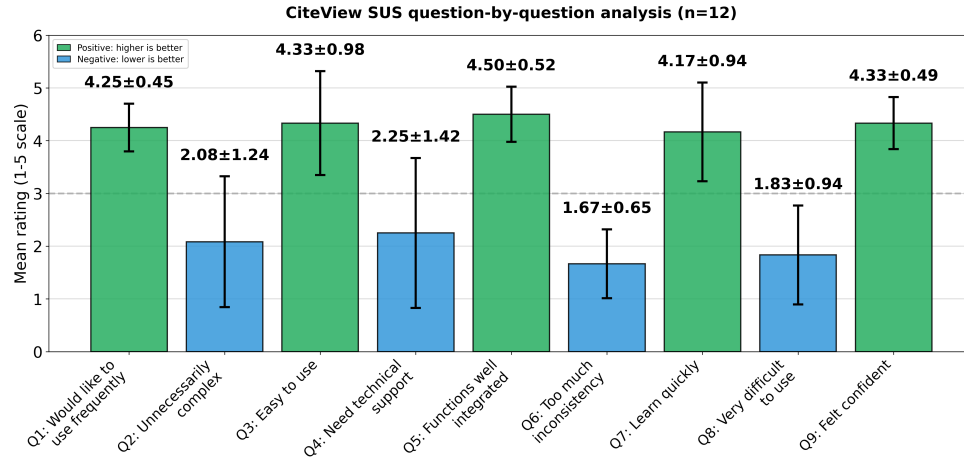


Fig. 8. Item-level SUS ratings for CiteView (n=12, 1-5 scale). Mean ratings with standard deviation error bars. Green bars represent positively worded items (odd-numbered: higher scores indicate better usability); blue bars represent negatively worded items (even-numbered: lower scores indicate better usability). The dashed line indicates the neutral midpoint (3.0). Results show strong agreement on positive items ($M=4.25-4.50$) and disagreement on negative items ($M=1.67-2.25$), indicating favorable usability perceptions.

6.3 Qualitative Themes

In the post-task interview, we asked participants to reflect on (i) whether CiteView met their expectations and why, (ii) The ease of navigation and any confusing elements, (iii) perceived accuracy and usefulness of the information shown in the sidebar, (iv) how the tool compared to their usual citation-following workflow, (v) which features they liked most/least, and (vi) what they would change or add in a future version. Across interviews, three recurring themes emerged: (1) reduced friction in accessing cited papers and contribution-level information, (2) strong preference for comparison- and insight-oriented views, and (3) a desire for a simpler, more consolidated interface with clearer visual explanations.

Theme 1: Reduced friction and faster access to contribution-level information. Participants consistently contrasted CiteView with traditional citation-following workflows, emphasizing speed and eliminated overhead. P1 explained the tool “made the information more readily available and easily accessible compared to looking for that individual PDF,” bringing contributions “instantaneously” and “in a short and concise way.” P4 estimated the efficiency gain was substantial: “It probably reduced it by 10X.” P2 similarly noted that with CiteView, “I can read more papers within a short period of time” compared to the overwhelming process of reading many papers traditionally. P3 highlighted the reduction in authentication and verification steps: without CiteView, they would need to “go on Google, figure out if I had access to that paper,” then “log into my student account, get access through that paper, and then confirm that it’s the paper that’s been cited,” whereas with CiteView, “I just have to click the citation.”

Theme 2: Preference for comparison, insights, and visualization-driven views. When asked about favorite features, participants most often highlighted views that supported interpretation and comparison rather than purely descriptive summaries. P1 stated “the results would be the most favourite because I can instantly understand what the cited paper did,” while P2 chose “the comparison, which I really like.” P3 emphasized “the graph and statistics tab because it provides

insights really quickly and highlighted the important graphs or figures.” Participants specifically valued the citation usage visualization, with P6 calling it “a very smart decision” because it shows “what this cited paper was used for the most, or cited in other papers for,” providing “good relevant data that people can use within their actual paper.” This preference aligned with interaction patterns where participants gravitated toward insight- and relationship-oriented views during citation-focused tasks.

Theme 3: Desire for simplification, clearer labeling, and stronger visual explanations. Although participants generally found the interface learnable, several described an initial learning curve and requested reducing verbosity. P1 acknowledged “there’s a little bit of a learning curve initially,” but emphasized it “becomes a lot more” manageable “once you understand what each tab does.” Multiple participants critiqued the Surface tab as too verbose or conceptually unclear. P4 stated they would “delete or modify” Surface because “I don’t understand what surface means,” recommending “just bullet points, bit shorter” instead of verbose paragraphs. P1 suggested to “merge the surface and the results” to achieve “less clutter on the UI,” while P3 wanted the first tab “more condensed” with highlights. P5 summarized the tension: the design was “amazingly designed,” but the main feedback was “getting lost with so many features.”

Participants also requested navigation improvements and better visual explanations. P6 proposed keyword search functionality: “If I could just search within CiteView, it would tell me a paragraph,” and “if there is any figure about it, it gives you an option to go to figures, if there are any results about it, it gives me an option to go to results.” P4 noted visualization clarity issues: “the graph didn’t have a legend that showed what the colours represented, so it was really hard to instantly get what is blue, what is gray.” Other requests included back navigation, clearer figure labeling to avoid ambiguity (P3: explicitly marking “teaser/main figure”), and improved full-screen viewing for visualizations.

Together, quantitative and qualitative results indicate CiteView supports effective citation exploration with low cognitive load, particularly through comparison and visualization features. Key improvement opportunities include consolidating overlapping views, adding keyword search navigation, and improving visual explainability through legends, labels, and interaction affordances.

7 Discussion

Our findings suggest that integrating contribution-focused citation analysis directly into the reading workflow can support faster sensemaking without substantially increasing cognitive load, particularly through comparison and visualization features.

Participants strongly preferred insight-oriented views over descriptive summaries, suggesting future designs should prioritize actionable information.

Remaining friction suggests future iterations should emphasize progressive disclosure and stronger visual scaffolding.

Based on our findings, we derive three design implications for citation exploration tools: (1) Prioritize Insight-Driven Views Over Descriptive Content. Participants strongly preferred tabs supporting interpretation (Results, Compare, Visualization) over descriptive summaries (Surface). P1 stated “the results would be the most favourite because I can instantly understand what the cited paper did,” while multiple participants critiqued Surface as “too verbose.” This suggests citation tools should foreground actionable information—what was contributed, how approaches differ, what evidence exists—rather than generic overviews. Designers might organize by analytical purpose (verify claims, compare methods, locate evidence) rather than document structure (abstract, methods, results), matching interface affordances to reader intent.

(2) Balance Feature Richness with Progressive Disclosure. Participants valued comprehensive capabilities (six dimensions, three figures, multiple result types) yet requested simplification. P5 noted the design was “amazingly designed” but gave feedback about “getting lost with so many features.” This tension reveals a key tradeoff: powerful tools require rich features, but complexity can overwhelm users. CiteView’s collapsible dimension cards exemplify progressive disclosure, readers scan six dimensions while selectively accessing details, but the four-tab structure requires upfront understanding of each tab’s purpose. Future designs should employ a hierarchical organization where high-level insights appear by default with clear pathways to deeper detail, maintaining analytical power while reducing initial cognitive load.

(3) Support Question-Driven Navigation Alongside Structured Exploration. Although structured tabs guided exploration, participants requested keyword search for direct access. P6 proposed: “If I could just search within CiteView, it would tell me a paragraph...if there is any figure about it, it gives you an option to go to figures.” This indicates citation tools should support both exploratory browsing (“What should I know?”) and targeted retrieval (“Does this report metric X?”). Effective tools accommodate both modes through curated overviews combined with in-tool search, recognizing that citation needs vary from comprehensive understanding to quick fact-checking.

Together, these implications suggest that effective citation tools should prioritize actionable insights over summaries, employ progressive disclosure to manage complexity, and support both exploratory and directed access patterns.

8 Limitations and Future Work

Our study has several practical limitations. First, CiteView depends on access to the full text of cited papers; however, not all references are openly available due to paywalls, missing PDFs, or incomplete metadata. As a result, some citations cannot be expanded into summaries, comparisons, or figure retrieval, which limits coverage and can introduce inconsistency across papers. Second, document quality and formatting remain a constraint: older or non-standard PDF layouts (e.g., scanned documents, uncommon templates, or poorly structured text layers) may reduce extraction reliability and may not be fully supported by the current pipeline, affecting citation detection, section parsing, and figure linking.

These limitations also motivate several directions for future work. To better support diverse reading contexts and accessibility needs, we plan to explore audio-based interaction (e.g., narrated citation summaries or voice-based playback of key contributions). We also aim to make citation exploration more interactive by adding keyword search and “jump-to” navigation across tabs, enabling users to quickly locate methods, results, or figures related to their question. Another direction is integrating an AI chat interface so users can ask targeted questions about a cited paper (or about how a citation is used in context) and receive grounded answers based on the retrieved content. Finally, consistent with participants’ requests for simplification, we plan to merge overlapping tabs and streamline the information architecture, keeping advanced views available through progressive disclosure while reducing perceived complexity for first-time users.

9 Conclusion

We presented CiteView, an interactive citation exploration tool that brings contribution-level context from referenced papers directly into the reading workflow. CiteView supports question-driven citation use by combining lightweight summaries with comparison, and relationship-oriented views, allowing readers to quickly understand what a cited work contributes and how it connects to the current paper without repeatedly switching tabs, searching externally, or navigating paywalls.

In a mixed-methods user study ($n = 12$), participants reported generally positive usability (SUS: $M = 71.9$, $SD = 13.2$) and low perceived workload on key effort-related dimensions, alongside high perceived time efficiency, ease of finding information, and overall helpfulness. Interview feedback further highlighted that CiteView reduced friction in accessing cited work, and that participants especially valued insight-driven views (e.g., results, comparison, and visualizations) for quickly answering citation-focused questions. At the same time, participants emphasized opportunities to simplify the interface, improve visual explainability, and better support in-tool navigation.

Overall, our findings suggest that citation interfaces should prioritize contribution-first access and insight-oriented representations while using progressive disclosure to balance power and learnability. We believe CiteView demonstrates that integrating contribution-level citation analysis directly into reading interfaces can support more fluid, question-driven scholarly exploration, transforming citation investigation from a disruptive detour into a natural part of the reading experience.

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